This dissertation, titled **"Quadcopter PID Controller Tuning for Vertical Motion,"** explores enhancing the vertical stability and performance of a quadcopter using Proportional-Integral-Derivative (PID) control and automated tuning methods.

1. Executive Summary and Scope

The primary goal of the research was to build a specialized simulation environment to optimize the **vertical motion** of a quadcopter. While quadcopters typically utilise six controllers for translational and rotational motion, this project isolates vertical movement to find the most efficient PID constants through systematic tuning.

- **Problem Definition:** Creating a high-fidelity simulator is complex, so the author utilised an open-source 3D quadcopter simulator by Peter Huang as a foundation.
- **Methodology:** The project was implemented entirely in **Python**, utilising libraries such as numpy, math, and matplotlib.
- **Key Achievement:** The implementation of a "step response" simulation, which triggers movement after a specific time to evaluate how the system handles sudden input changes.

2. Technical Implementation

The author transitioned a default PD (Proportional-Derivative) controller into a full **PID controller** by adding an integral (I) component to eliminate steady-state error.

Core Components

- **Integral Error & Clipping:** To prevent "integral windup" (excessive error accumulation), the integral term was clipped before being added to the final control signal.
- **Yaw Normalization:** A specific logic was added to normalize yaw error within the range of $[-\pi, \pi]$ for smooth angular calculations.
- **Anti-Windup Method:** This advanced feature checks physical thrust limits; if the system is saturated, the integral error stops updating to maintain stability.

3. Tuning Methods & Performance Metrics

The thesis focuses on two primary methods for finding optimum PID gains (K_p, K_i, K_d):

A. Ziegler-Nichols Method

A heuristic method based on finding "consistent oscillations".

- Set K_i and K_d to zero and gradually increase K_p until a stable oscillation (the "ultimate gain") is reached.
- Calculates the **ultimate period** based on zero-crossings in the 3D simulation.

B. Grid Search Tuning

A simpler but more computationally expensive search method.

- Iterates through specified ranges of PID constant combinations.
- Selects the combination that results in the lowest **Mean Squared Error (MSE)**.

Evaluated Metrics

The performance of the tuned constants was compared against untuned defaults using a two-stage simulation environment:

- **Rise Time:** Time to go from 10% to 90% of the target distance.
- **Settling Time:** Time to stay within 2% of the final target height.
- **Overshoot:** The percentage by which the quadcopter surpasses the target point.
- **Steady-State Error (SSE):** The final difference between actual and desired height.

4. Results and Conclusion

The study found that the "better" method often depends on the gain ranges used:

Metric	Winner	Reasoning
MSE & Overshoot	**Grid Search**	Specifically designed to minimize squared error.
Settling Time & SSE	**Ziegler-Nichols**	Better at reaching and maintaining the exact target height.
General Performance	**PD Controller**	Interestingly, the author noted the default PD controller was slightly smoother than the basic PID before advanced tuning.

Final Outcome: The integration of the **Anti-Windup** method significantly improved performance when paired with grid-search tuned gains, leading to much lower overshoot (reduced from ~43% to ~1.6% in some tests) and faster settling times. The project concludes that these vertical-specific improvements can be adapted for more complex 3D flight tasks in other robotics applications.